

Retail Inventory & Demand Dashboard

Inventory health, risk, and demand forecasting — powered by Python + Power BI

10M

Total Units Sold

2.16

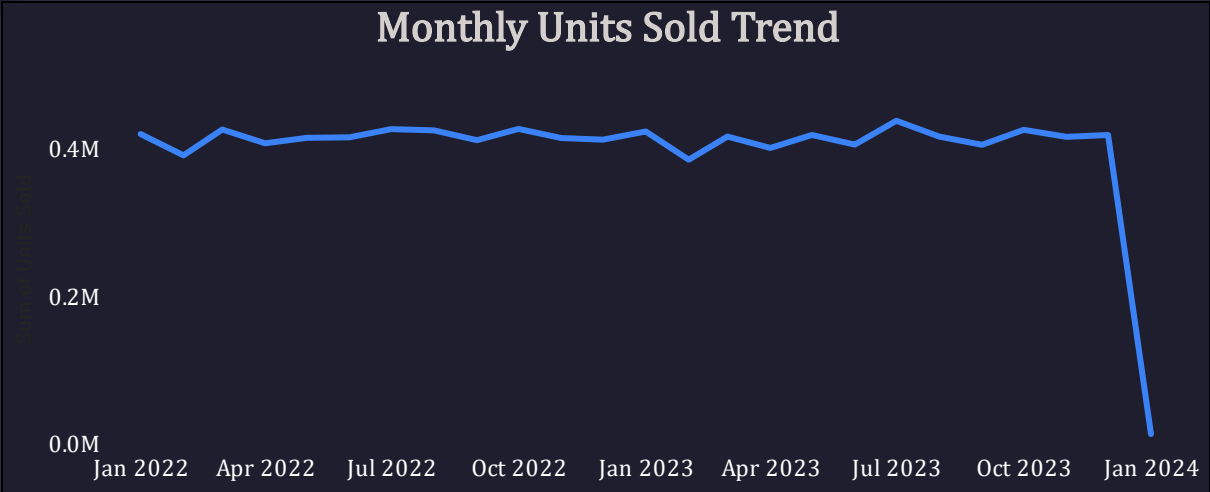
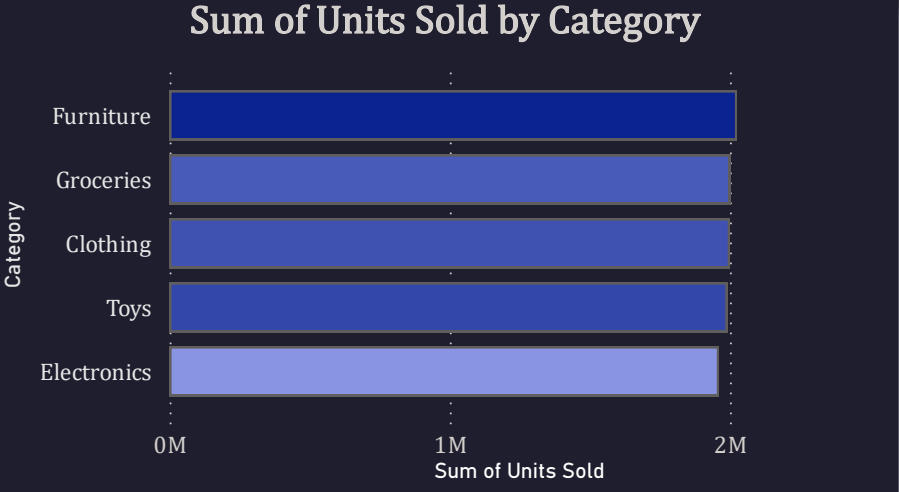
Avg Stock Cover Days

0.25

% Low Stock

0.31

Avg Sell Through Rate



Sales are evenly distributed across categories/regions — no dominant seller. Discount showed no impact on units sold ($r=0.003$)

Inventory Risk & Reorder Analysis

Identifying which products are running low and need restocking

0.47

% Needs Reorder

Urgent Restock Table

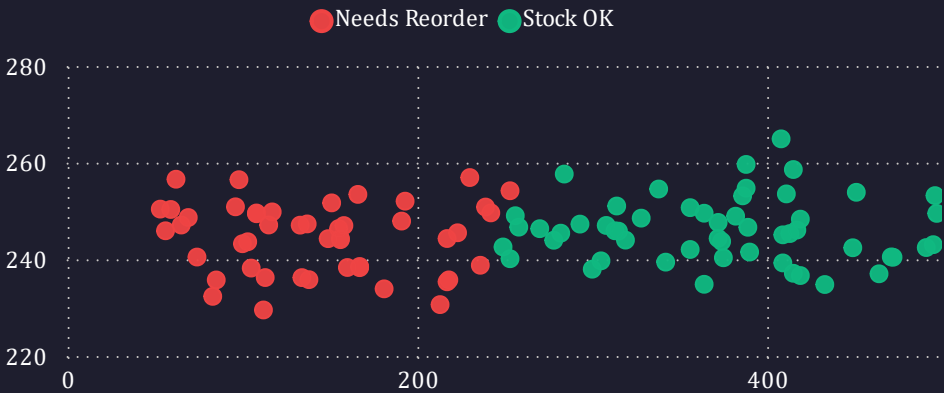
Store ID	Product ID	Current_Inventory	Reorder_Point
S002	P0007	53	250.39
S004	P0015	56	245.91
S005	P0007	59	250.31
S002	P0009	62	256.57
S004	P0019	65	247.10
S001	P0003	69	248.65
S001	P0016	74	240.47
S004	P0018	83	232.34
S004	P0012	85	235.73
S005	P0016	96	250.87
S005	P0003	98	256.48
S004	P0017	100	243.23
S003	P0008	103	243.64
S004	P0004	105	238.19
S002	P0007	109	240.50

Reorder point = demand × lead time + safety stock.
Assumptions revised after checking against real inventory scale. 47% of store-product pairs are below reorder point.

Low-stock rates are consistently ~25% across all categories and regions — no localized risk pattern.

Category	East	North	South	West
▼				
Toys	0.24	0.24	0.26	0.25
Groceries	0.25	0.26	0.24	0.27
Furniture	0.25	0.24	0.24	0.25
Electronics	0.25	0.25	0.26	0.25
Clothing	0.26	0.25	0.25	0.25

Inventory Risk Quadrant -Current Stock vs Reorder Point

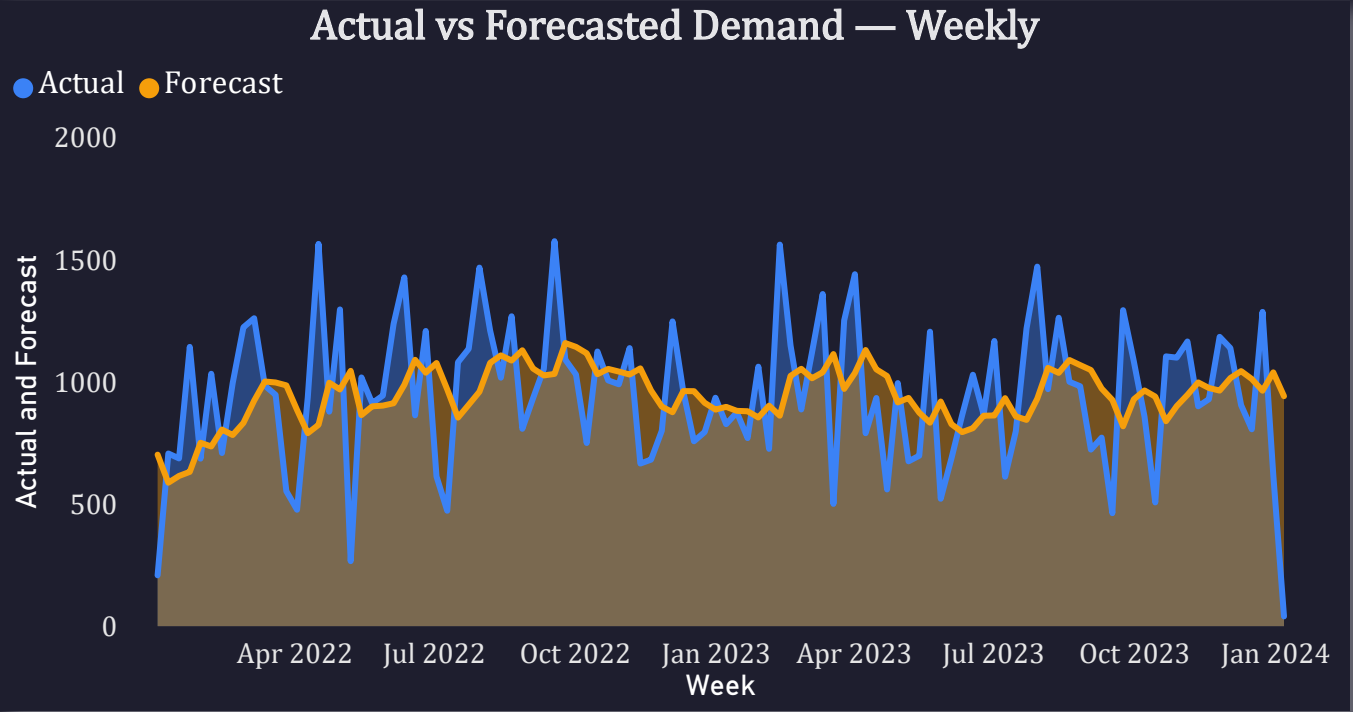
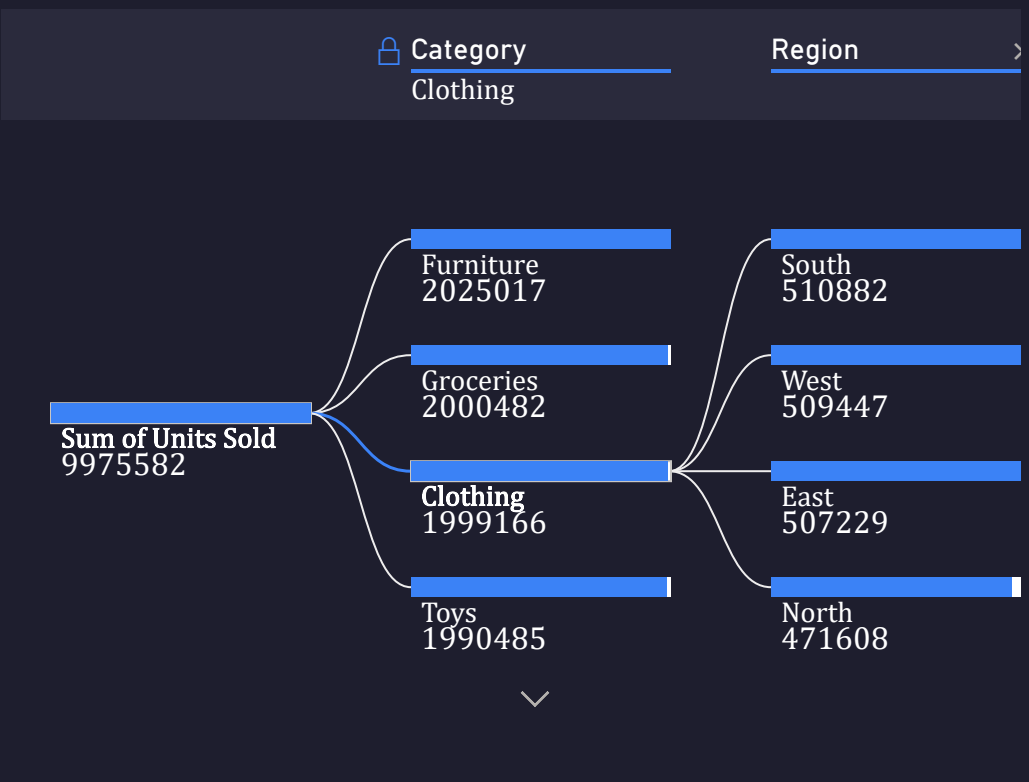


Demand Forecasting & Model Performance

Testing whether historical demand patterns can predict future sales

26%
Weekly Forecast Error

65%
Daily Forecast Error



Daily forecasts are noisy (65% error); weekly aggregation smooths this to 26% — weekly is reliable for planning, daily is not